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Sorting Properties.

1) In place :

An algorithm is in place if it requires O(1) or (O(log n) for recursive call stacks)

If it requires more memory, it is NOT in place.

e.g

- Selection, Bubble, insertion sort are all In place sorts.

- Quick sort is also inplace because it does not allocate any new array.

Merge Sort is NOT Inplace.

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2) Stability:

A sorting algorithm is stable if it preserves the relative order of elements that have equal keys (values).

Understanding through example:

consider array: $(1, 2, 5, 3_1, 8, 3_2)$

To differentiate both 3's.

If result:

~~(1, 2, 5, 3, 8)~~

$(1, 2, 3_1, 3_2, 5, 8.)$ Stable

If:

$(1, 2, 3_2, 3_1, 5, 8.)$ Unstable.

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